Mini Project

Exploratory Data Analysis on India's COVID-19 Data

State/Province-wise trends, recovery rates, case severity and temporal patterns

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Dataset:Covid-19.csv

Analysis period: 21 January 2020 to 26 January 2020

Tools: Python, Pandas, NumPy, Matplotlib and Seaborn

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1 Introduction

The COVID-19 outbreak generated a large amount of health-related data. Analysing such data helps in understanding how cases were distributed across locations and how reported outcomes changed over time. Exploratory Data Analysis provides a practical way to inspect this information before applying more advanced statistical or machine-learning methods.

This project focuses on an early COVID-19 summary dataset supplied for the mini project. The analysis is designed around the concepts of data loading, inspection, cleaning, transformation, grouping, statistical calculation and visualization.

A key point about the supplied file is that it is an early global dataset, not an India-only or vaccination dataset. Therefore, the report does not invent India-specific vaccination results. Instead, it analyses exactly the variables available in the supplied CSV file.

Problem Statement
To explore the supplied early COVID-19 dataset and identify meaningful patterns in confirmed cases, suspected cases, recoveries, deaths and active cases across countries and provinces/states using Python-based EDA techniques.

Significance of the Study
The project demonstrates how raw public-health data can be converted into understandable information through systematic preprocessing and visualization. It also demonstrates the importance of checking the scope and limitations of a dataset before drawing conclusions.

2 Dataset Description

The analysis uses the supplied file **Covid-19.csv**. The dataset contains 368 records and covers observations from 21 January 2020 to 26 January 2020.

Original Variables

Variable	Description
Province/State	Province or state associated with the observation.
Country	Country label associated with the observation.
Date last updated	Date and time at which the observation was recorded/updated.
Confirmed	Number of confirmed COVID-19 cases.
Suspected	Number of suspected cases.
Recovered	Number of reported recovered cases.
Deaths	Number of reported deaths.

Derived Variables

Derived Feature	Calculation / Purpose
Active Cases	Confirmed − Recovered − Deaths
Resolved Cases	Recovered + Deaths
Recovery Rate	(Recovered / Confirmed) × 100
Death Rate	(Deaths / Confirmed) × 100
Suspected Rate	(Suspected / Confirmed) × 100
Year, Month, Day	Extracted from the date field.
Daily New Cases	Change in confirmed cases within each country over time.

3 Objectives

The major objectives of the project are:

1. To load and inspect the supplied COVID-19 dataset using Pandas.
2. To identify and handle missing values and incorrect data types.
3. To create meaningful derived features such as Active Cases and Recovery Rate.
4. To analyse country-wise and province/state-wise case patterns.
5. To study confirmed cases, recoveries and deaths over time.
6. To compare recovery and death rates across locations.
7. To examine relationships among major numerical COVID-19 variables.
8. To communicate the results using clear Matplotlib and Seaborn visualizations.
9. To document the findings, limitations and possible future extensions.

4 Data Cleaning and Feature Engineering

Missing Values

The dataset contains missing values in some fields. Missing province/state labels are replaced with 'Unknown', while missing numerical observations are treated as zero for the calculations used in this project.

Date Conversion

The 'Date last updated' field is converted into datetime format. This makes it possible to extract Year, Month and Day and to create time-based visualizations.

Active Cases

Active Cases = Confirmed − Recovered − Deaths. This derived variable provides a simple estimate of reported cases that had not yet been recorded as recovered or deceased in the snapshot.

Recovery and Death Rates

Recovery Rate = (Recovered / Confirmed) × 100

Death Rate = (Deaths / Confirmed) × 100

These are descriptive ratios. Because the dataset represents only the early stage of the outbreak, the values should not be interpreted as final clinical outcome rates.

Daily New Cases

Daily New Cases are calculated from the change in confirmed cases within each country over time.

Negative differences caused by reporting revisions are clipped to zero for the project visualization.

5 Exploratory Data Analysis and Visualization

The following charts present the main visual findings from the project. Each figure is followed by an interpretation based on the supplied data.

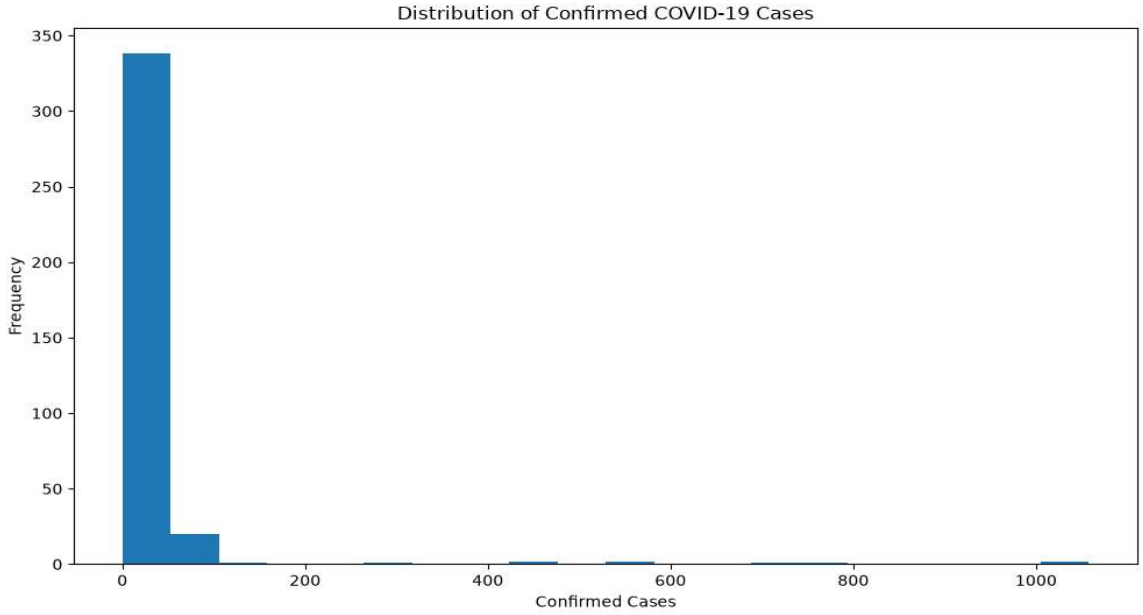


Figure 1. Confirmed cases Covid-19 over the available period.

Interpretation: *Confirmed cases increase substantially across the short observation window.*

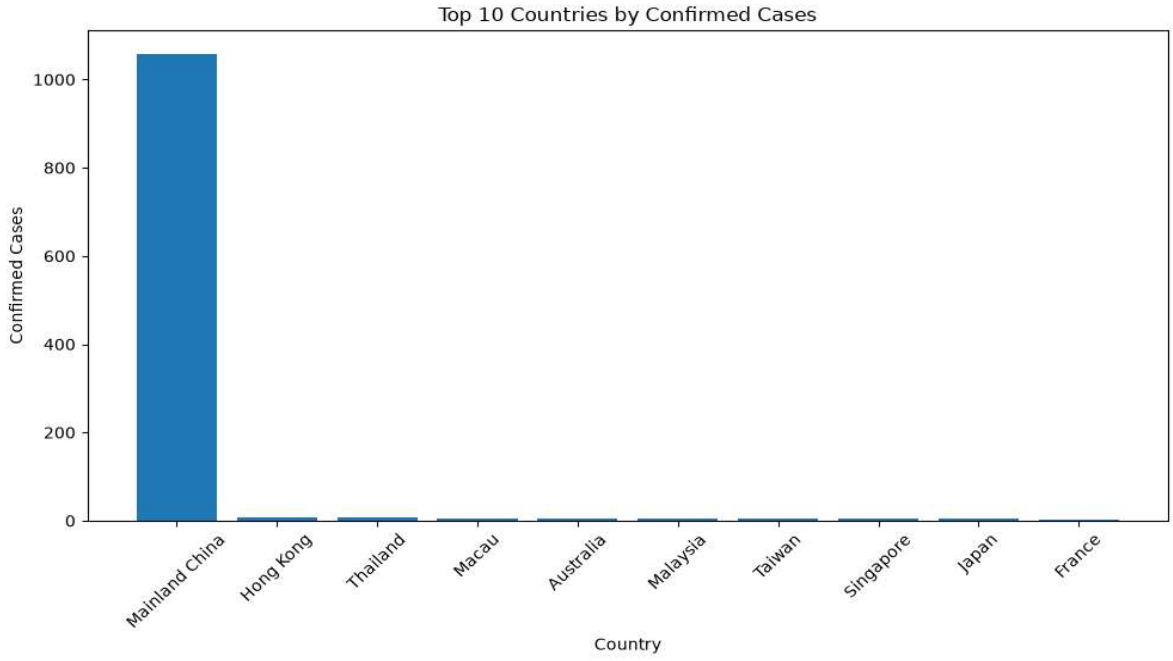


Figure 2. Top 10 countries by confirmed cases at the latest available snapshot.

Interpretation: *Mainland China has a much larger reported case count than the other country labels in the latest snapshot. The distribution demonstrates strong geographical concentration during this early period.*

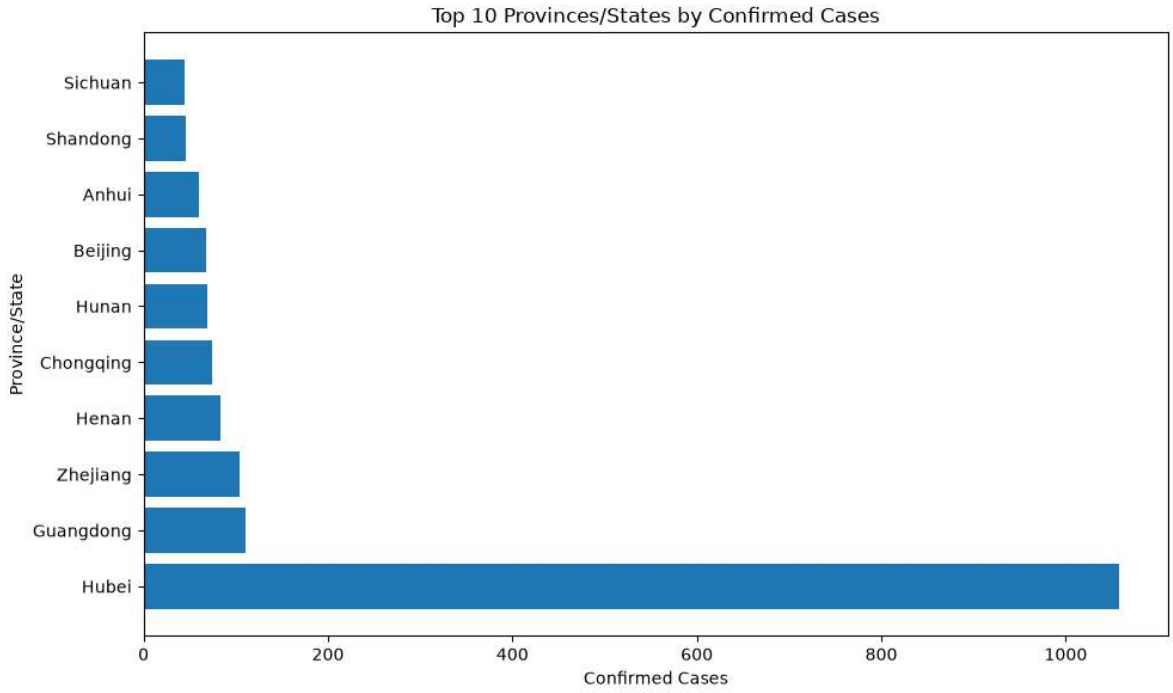


Figure 3. Top 10 provinces/states by confirmed cases.

Interpretation: *Hubei is the largest province/state-level location by confirmed cases, followed by other Chinese provinces such as Guangdong, Zhejiang and Henan.*

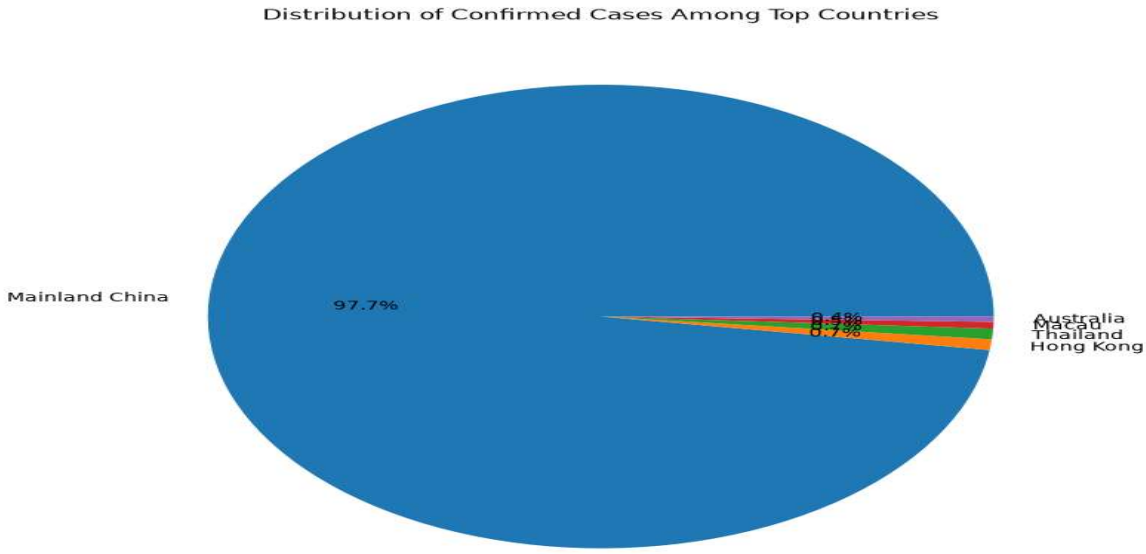


Figure 4. Confirmed-case share of the five leading country labels.

Interpretation: *The pie chart reinforces the concentration seen in the ranking chart. A large share of the reported cases belongs to the leading country labels, especially Mainland China.*

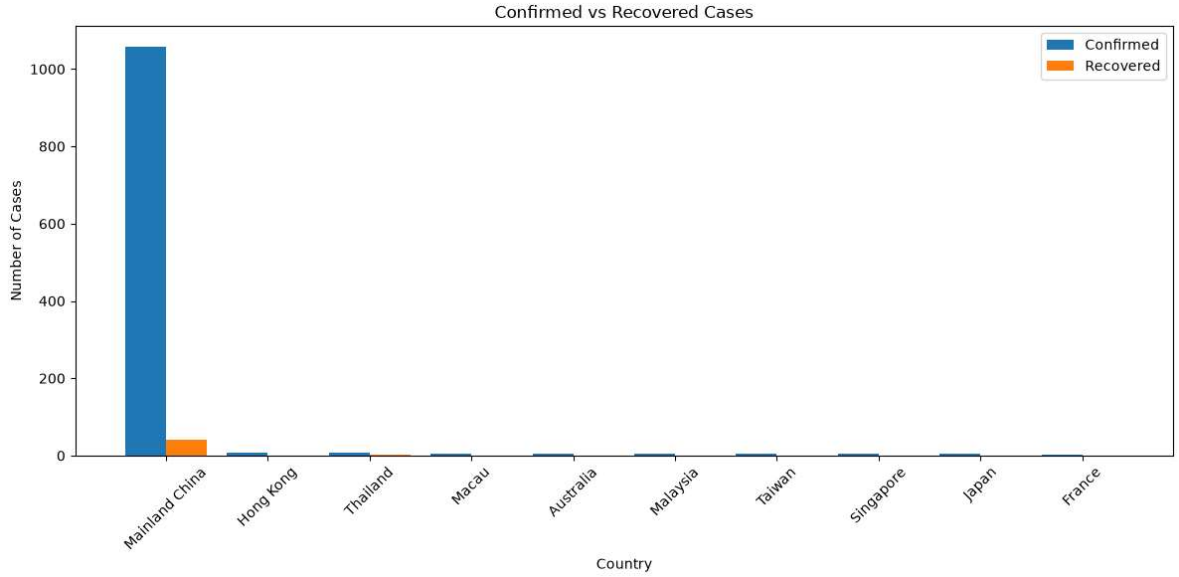


Figure 5. Confirmed Vs Recovery pages.

Interpretation: *Confirmed percentages can vary widely when Recovered Cases are small*

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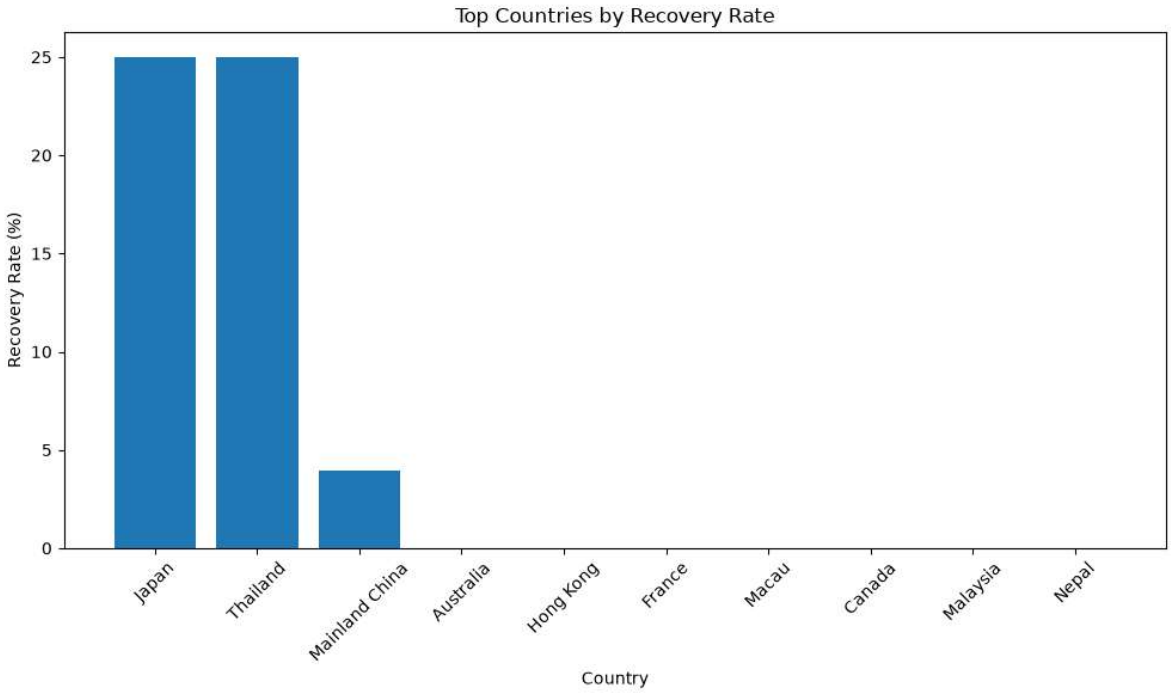


Figure 6. Top Countries by Recovery Rate

Interpretation: *The recovery rate changes as reported recoveries increase. The short time span and early reporting conditions make this an early-stage descriptive measure.*

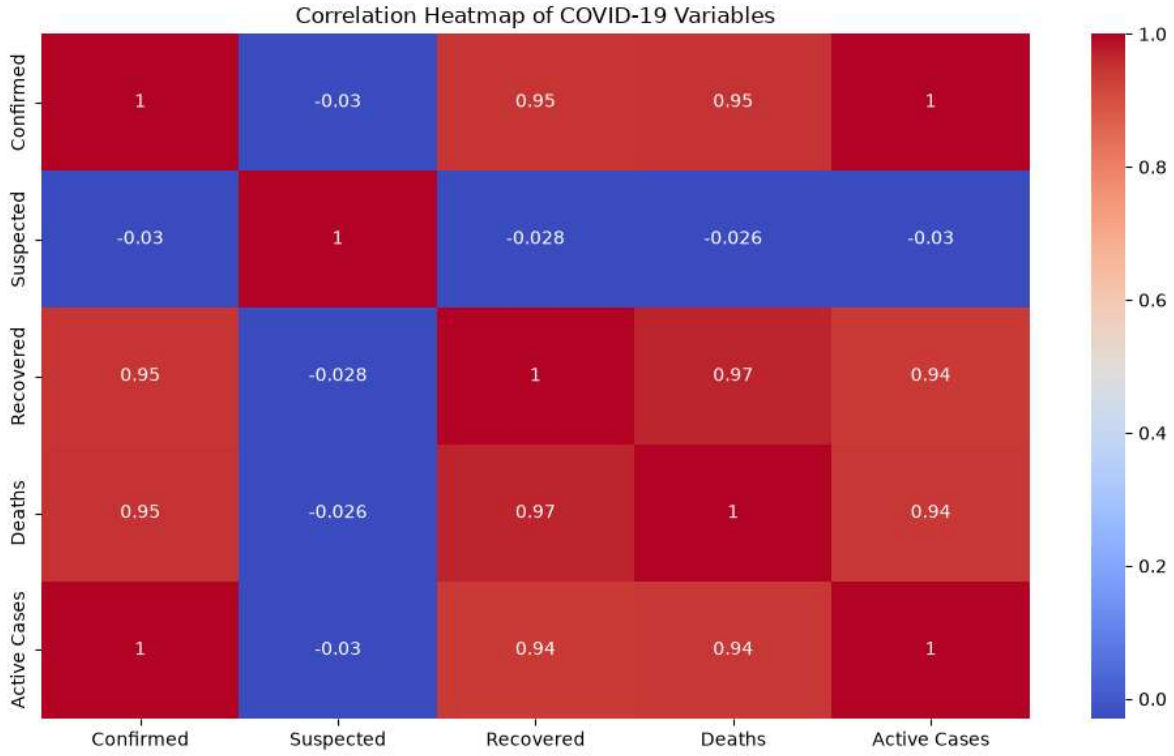


Figure 7. Correlation heatmap of the COVID-19 variables.

Interpretation: *Confirmed, recovered, deaths and active cases show strong relationships because they are cumulative or mathematically derived measures. The heatmap is useful for understanding which variables move together in the supplied dataset.*

6 Key Findings

The main observations obtained from the exploratory analysis are summarized below:

1. The dataset is an early global COVID-19 snapshot covering 21 January 2020 to 26 January 2020.
2. The supplied file contains 368 records and is dominated by observations associated with China and its provinces.
3. Mainland China has the largest country-level confirmed-case count in the latest snapshot.
4. Hubei is the most affected province/state-level location in the supplied data.
5. Confirmed cases increased rapidly over the short observation window.
6. Recovery and death rates differ across locations, but the early timing and small denominators make these percentages unstable.
7. Active Cases provide a useful derived measure for distinguishing total confirmed cases from cases not yet recorded as resolved.
8. The correlation heatmap demonstrates strong relationships among cumulative and derived COVID-19 variables.

The dataset does not contain vaccination variables; therefore, vaccination progress cannot be calculated from this file alone.

7 Future Scope & Conclusion

Future Scope

The project can be extended considerably if a larger and more complete dataset is available

1. Use a longer COVID-19 time-series dataset to study the complete outbreak trajectory.
2. Standardize country and province/state names before performing aggregation.
3. Add population data and calculate confirmed cases and deaths per 100,000 people.
4. Add vaccination data to analyse vaccination coverage and its relationship with later cases.
5. Use moving averages to smooth daily fluctuations and reporting noise.
6. Apply forecasting or time-series methods to study future trends.
7. Build an interactive dashboard using tools such as Streamlit or Power BI.

Conclusion

This mini project demonstrates a complete Exploratory Data Analysis workflow on the supplied early COVID-19 dataset. The project begins with data loading and inspection, followed by missing-value handling, date conversion and feature engineering. Meaningful derived variables such as Active Cases, Recovery Rate and Death Rate make the dataset more suitable for analysis.

The visual analysis shows that the early outbreak represented in the dataset was highly concentrated geographically, with Mainland China and Hubei accounting for a major share of reported cases. Confirmed cases increased rapidly during the short observation period, while recovery and death measures were still relatively limited.

The project also highlights an important data-science principle: conclusions must match the scope of the data. Since the supplied dataset does not contain vaccination information and covers only the early days of the outbreak, it would be inappropriate to claim vaccination trends or complete-pandemic conclusions.

Overall, the project provides practical experience in Python-based data cleaning, feature engineering, grouping, visualization and interpretation, and establishes a foundation for more advanced COVID-19 analysis using a larger and more complete dataset.